SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	AI&DS	**Date:**	10/10/2026

Code of Conduct

I Roshan / 192524085 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Roshan R

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

Question 1: Answer

(a) Identify Correct and Incorrect Statements

Statement	Correct / Incorrect	Erroneous Phrase / Claim
S1	Incorrect	The Turing Test does not check whether internal reasoning always follows the laws of logic. It evaluates intelligent behaviour through interaction. It represents the acting humanly approach, not thinking rationally.
S2	Incorrect	AI winters did not occur in the 1950s before the Dartmouth workshop. The 1956 Dartmouth workshop helped establish AI as a field and introduced the term Artificial Intelligence.
S3	Incorrect	A rational agent does not always achieve the best possible outcome and does not know the result of every action in advance. Rationality depends on available information and expected performance.
S4	Incorrect	The PEAS components are incorrectly assigned. Steering, accelerator, brake and horn are actuators, while safe, fast, legal and comfortable travel are performance measures.
S5	Incorrect	Chess without a clock is fully observable, deterministic, sequential, static and discrete. It is not partially observable, stochastic or episodic.
S6	Correct	No change. A model-based reflex agent maintains an internal state and uses knowledge of environmental changes and action effects.

(b) Root Cause Analysis

S1: The Turing Test is confused with the thinking rationally approach. The Turing Test evaluates observable behaviour rather than the internal reasoning process.
S2: The chronology of AI development is misunderstood. The establishment of AI in 1956 is confused with later periods of reduced funding and interest known as AI winters.
S3: Rationality is confused with omniscience. A rational agent makes decisions using available percepts, knowledge and possible actions. It cannot predict every unforeseen event.
S4: The PEAS framework is misunderstood. Performance measures evaluate success, the environment contains external conditions, actuators perform actions, and sensors collect information.

S5: The properties of task environments are incorrectly classified. Chess provides complete information about the board, has deterministic rules, involves sequential decisions, and uses discrete states and actions.

S6: No misconception is present. The statement correctly describes the internal state and environmental model of a model-based reflex agent.

(c) Corrected Statements and Justifications

Statement	Corrected Statement	Justification
S1	The Turing Test evaluates whether a machine's conversational behaviour is indistinguishable from a human to an evaluator. It represents the acting humanly approach.	The test focuses on observable behaviour rather than internal logical reasoning.
S2	The 1956 Dartmouth workshop helped establish AI as a field and introduced the term Artificial Intelligence. AI winters occurred later.	The chronology distinguishes the establishment of AI from subsequent funding downturns.
S3	A rational agent selects the action expected to maximize its performance measure based on available percepts, knowledge and actions. The taxi can be rational even if an unforeseeable landslide blocks its route.	Rationality is evaluated using information available when the decision is made.
S4	Performance measure: safe, fast, legal and comfortable travel. Environment: roads, traffic, pedestrians and weather. Actuators: steering, accelerator, brake and horn. Sensors: cameras, GPS and speedometer.	Each PEAS component is correctly classified.
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete. It is also a multi-agent, adversarial environment.	The board is visible, the rules determine outcomes, and each move affects subsequent decisions.
S6	No change. A model-based reflex agent maintains an internal state updated using a model of environmental changes and action effects.	The internal state helps the agent make decisions when the current percept does not reveal the complete situation.

(d) Application to the Self-Driving Taxi

(i) Was the taxi's decision rational?

The taxi's decision was rational if it selected the highway route using all available traffic reports and relevant knowledge.

The four factors determining rationality are:

1. **Performance measure:** Safe, timely and legal transportation.
2. **Percept sequence:** Traffic reports and sensor information received before choosing the route.
3. **Prior knowledge:** Knowledge of roads, traffic conditions and the environment.
4. **Available actions:** The possible routes and driving actions available to the taxi.

The unforeseeable landslide does not make the original decision irrational because the taxi could not have known about the event in advance.

(ii) Task-environment properties that prevent perfect foresight

The taxi operates in a partially unpredictable and dynamic environment. Unexpected events, such as landslides, can occur after the decision is made. The environment is also partially observable because sensors and traffic reports cannot reveal every future condition.

(iii) Simple reflex taxi versus model-based reflex taxi

A simple reflex taxi may continue driving because its current camera percept shows a clear road, even though it previously encountered a warning about a road closure.

A model-based reflex taxi maintains an internal state containing relevant information from previous precepts. It uses a model of how the environment changes and how its actions affect the environment to make better decisions.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		

Component / Note	Correct / Incorrect	What is wrong
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 2: Answer

(a) Identify Errors in the Formulation

Component / Note	Correct / Incorrect	What Is Wrong?
State representation	Incorrect	The state representation omits the boat's location.
Initial state	Incorrect	The initial state must include the boat's location: (3, 3, L).
Actions	Correct	The listed passenger combinations represent the possible boat loads. Each action must be feasible and must preserve safety on both banks.
Goal test	Incorrect	M = C is not the goal. The goal is to move everyone to the right bank with the boat there.
Legality check	Incorrect	The student checks only the left bank. Safety must be checked on both banks.
Path cost	Incorrect	Counting the number of people carried does not minimize crossings. Each crossing should have a cost of 1.
Note N1	Incorrect	The correct order is goal formulation → problem formulation → search → execution.
Note N2	Incorrect	The state space contains 32 possible states before legality filtering, not 16. There are 20 legal states under the stated safety rules.

(b) Root Cause Analysis

State representation and initial state: The boat's location is essential because it determines which actions are available. Omitting it prevents the state from fully describing the problem.
Actions: Actions must satisfy boat capacity, passenger availability and bank-safety constraints.
Goal test: The student confuses a relationship between missionary and cannibal counts with the actual objective of moving everyone to the right bank.
Legality check: The student checks only one bank even though the safety condition applies to both banks.

Path cost: The selected cost does not match the objective of minimizing the number of crossings.

Note N1: The student reverses the standard problem-solving sequence by searching before formulating the goal and problem.

Note N2: The student omits the boat's position when counting states. There are four possible missionary counts, four possible cannibal counts and two possible boat positions, giving $4 \times 4 \times 2 = 32$ possible states.

(c) Corrected Formulation and Justification

Component / Note	**Corrected Formulation**	**Justification**
State representation	(M, C, B), where M and C are the numbers of missionaries and cannibals on the left bank and B is the boat's location, L or R.	It includes all information required to determine available actions and successor states.
Initial state	(3, 3, L)	Everyone and the boat begin on the left bank.
Actions	Move 1M, 2M, 1C, 2C or 1M + 1C across the river, provided the passengers are on the boat's current bank and both banks remain safe.	It respects boat capacity and safety constraints.
Goal test	(0, 0, R)	All missionaries and cannibals are on the right bank, and the boat is also on the right.
Legality check	On each bank, missionaries must not be outnumbered by cannibals unless there are no missionaries on that bank. Also, $0 \leq M, C \leq 3$.	Safety must be maintained on both banks, and population counts must be valid.
Path cost	Cost = 1 per crossing.	This directly minimizes the number of crossings.
Note N1	Goal formulation → problem formulation → search → execution.	The goal and problem must be defined before searching for a solution.
Note N2	There are 32 possible states, of which 20 are legal.	The boat position doubles the 16 combinations of missionary and cannibal counts.

(d) Apply the Corrected Formulation

(i) Boat Loads Available from the Initial State

The initial state is (3, 3, L).

Boat Load	**Resulting State (M, C, Boat)**	**Legal? Reason**
1M	(2, 3, R)	Illegal. On the left bank, 2 missionaries would be outnumbered by 3 cannibals.
2M	(1, 3, R)	Illegal. On the left bank, 1 missionary would be outnumbered by 3 cannibals.
1C	(3, 2, R)	Legal. The left bank has 3 missionaries and 2 cannibals. The right bank has no missionaries, so it is safe.
2C	(3, 1, R)	Legal. The left bank has 3 missionaries and 1 cannibal. The right bank has no missionaries, so it is safe.
1M + 1C	(2, 2, R)	Legal. The left bank has equal numbers of missionaries and cannibals, and the right bank is safe.

(ii) Give One State Accepted by the Student's Legality Check but Actually Illegal

Answer: (2, 1, R).

The student's check accepts this state because $1 \leq 2$ on the left bank. However, the right bank contains 1 missionary and 2 cannibals, so the missionaries are outnumbered. Therefore, the state is illegal.

(iii) State the Cost of an Optimal Solution

The optimal solution requires **11 crossings**.

Since each crossing has a cost of 1, the total optimal path cost is **11**.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 3: Answer

(a) Identify Errors in the Traces and Remarks

No.	Where	What Is Wrong?
1	BFS, iteration 3	The queue should be [A, B, E, G], not [B, E, G, A]. The existing queue entry A must remain ahead of newly generated nodes.
2	BFS trace	The correct dequeue order is S, C, D, A, B, E, G. The goal is tested when G is removed from the frontier.
3	Remark R1	An explored set is necessary in graph search to prevent repeated expansion of already explored nodes.
4	DFS, iteration 1	The stack should be [C, D], not [D, C], because the alphabetically first child must be popped first.
5	DFS trace and result	The incorrect stack order changes the search path. The corrected DFS path is S–C–A–B–G, with cost 14.
6	UCS, iteration 4	The path through E gives G a cost of 8, which is cheaper than the existing cost of 9. The frontier entry must be updated.
7	Remarks R2 and R3	BFS and DFS are not necessarily cost-optimal in a weighted graph. UCS finds the least-cost path. DFS is not automatically the best strategy because it removes fewer nodes.

(b) Root Cause Analysis

1. BFS queue order: BFS follows FIFO order. Existing frontier nodes remain ahead of newly generated nodes.
2. BFS goal test: The goal must be tested when a node is removed from the frontier, as specified in the question.
3. Explored set: Graph search must detect repeated states to prevent redundant expansions and problems caused by cycles.
4. DFS stack order: A stack follows LIFO order. The alphabetically first child must be placed at the top so that it is popped first.
5. DFS trace: An incorrect stack order changes the traversal order and the resulting solution path.

6. UCS priority queue: UCS selects the node with the smallest cumulative path cost. A frontier entry must be updated when a cheaper path is discovered.

7. Optimality: BFS minimizes the number of edges when step costs are equal. DFS does not guarantee an optimal path. UCS finds an optimal path when edge costs are non-negative.

(c) Corrected Traces and Justifications

(i) BFS: Corrected Queue After Iteration 3

Queue after iteration 3: [A, B, E, G]

Order of dequeued nodes:

S → C → D → A → B → E → G

The queue follows FIFO order. Existing entries remain ahead of newly generated nodes, and the goal is tested when G is removed from the queue.

(ii) Corrected DFS Trace

The stack's top is on the left.

Iteration	Node Popped	Stack After Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

Justification: DFS uses a stack, so the top node is processed first. The corrected order follows the alphabetical-first child convention. The goal G is reached through the path S–C–A–B–G.

(iii) Corrected UCS Trace

Iteration	Node Expanded	g(n)	Priority Queue After Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), G:9(D), A:9(C)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	—

Justification: When E is expanded, the path S–D–E–G has cost 3 + 2 + 3 = 8. This is cheaper than the existing path to G with cost 9. Therefore, UCS updates G's cost to 8 and removes G from the frontier next.

(e) Apply the Corrected Traces

(i) Comparison of BFS, DFS and UCS

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S–D–G	9	7	No. BFS minimizes the number of edges, not weighted path cost.
DFS	S–C–A–B–G	14	5	No. DFS does not guarantee the least-cost path.
UCS	S–D–E–G	8	6	Yes. UCS finds the least-cost path.

(ii) Why Is an Explored Set Needed in Graph Search?

When D is expanded during BFS, some of its neighbours may already have been explored or added to the frontier through another path.

The explored set prevents already expanded nodes from being expanded again. Checking the frontier also prevents duplicate queue entries.

Therefore, an explored set reduces redundant work and helps graph search terminate on graphs containing cycles.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	

No.	Root cause (concept or rule misapplied)
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Question 4: Answer

(a) Identify Errors in the Tables and Conclusions

No.	Where	What Is Wrong?
1	Task A, Step 1	No error. After Suck, the belief state {S5, S7} is correct.
2	Task A, Step 3	After Suck, both S6 and S8 transition to S8. The student's final belief state is incorrect.
3	Task B, Step 0	After percept [Left, Dirty], the correct belief state is {S1, S3}, not {S1, S2, S3, S6}.
4	Task B, Step 1	After Suck, the predicted belief state should be {S5, S7}, not {S4, S5, S7, S8}.
5	Task B, Step 2	After Right, the predicted belief state should be {S6, S8}, not {S6, S7}.
6	Conclusion C1	The claim that sensorless problems cannot be solved is incorrect. The plan [Suck, Right, Suck] guarantees that both squares become clean.
7	Conclusions C2 and C3	A singleton belief state identifies the physical state but does not necessarily mean the goal is achieved. Also, 8 physical states produce $2^8 = 256$ possible belief states, not 8.

(b) Root Cause Analysis

1. Task A, Step 1: No error is present. The Suck action correctly cleans the left square in every possible state.

2. Task A, Step 3: The student fails to apply the transition rule for Suck. Both possible states become S8 after cleaning the right square.

3. Task B, Step 0: The student does not filter the belief state using both parts of the percept: the agent's location and the current square's condition.

4. Task B, Step 1: The student introduces states that are not successors of the previous belief state. Each possible state must be transformed using the same action.

5. Task B, Step 2: The student incorrectly applies the Right action. It changes the agent's location while preserving the dirt conditions.

6. Conclusion C1: The student assumes that a belief state containing multiple possibilities prevents a guaranteed solution. However, a suitable action sequence can make every possible state satisfy the goal.

7. Conclusions C2 and C3: The student confuses knowing the exact physical state with achieving the goal and misunderstands the size of the belief-state space.

(c) Corrected Tables and Conclusions

(i) Task A: Corrected Belief States

Step	Action	Belief State After Action
0	Start	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

Justification: Suck cleans the square where the agent is located. After moving Right, the agent cleans the right square. Both possible states then become S8, where both squares are clean.

(ii) Task B: Corrected Belief States

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

Justification: The predicted belief state is obtained by applying the action to every state in the current belief state. The updated belief state retains only the states consistent with the received percept.

(iii) Corrected Conclusions C1 and C2

C1:

The sensorless plan [Suck, Right, Suck] guarantees the goal for the given initial belief state. The final belief state is {S8}, representing (R, C, C). Therefore, both squares are clean.

C2:

The final belief state is {S6}, so the agent knows the exact physical state. However, the goal has not been achieved because S6 represents (R, C, D), where the right square remains dirty.

(d) Apply the Corrected Concepts

(i) How Many Different Belief States Are Possible?

There are $\mathbf{2^8 = 256}$ **possible belief states** when there are eight physical states.

Each physical state can either be included in or excluded from a belief state. Therefore, there are two choices for each of the eight states.

Belief-state search can be harder than physical-state search because the number of possible belief states grows exponentially with the number of physical states.

(ii) Apply the Plan [Right, Suck]

Initial belief state:

{S1, S3, S5, S7}

Step	Action	Belief State After Action
0	Start	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}

Is it a solution? No.

After Right, the agent moves to the right square in every possible state. Suck cleans the right square, but the left square may still be dirty.

The final belief state contains S4, representing (R, D, C), where the left square is dirty. Therefore, the plan does not guarantee that both squares are clean.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____